

Zabbix Post-Deployment Guide

This guide covers everything needed after the Zabbix server is up: securing the server, installing agents on hosts, adding hosts to monitoring, applying templates, and setting up alerting.

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1. First Login & Security

Change the Default Admin Password

1. Log in at `http://<zabbix-ip>/` with **Admin / zabbix**
2. Click the user icon (top right) → **Profile**
3. Click **Change password**
4. Set a strong password and save

Rename the Default Admin User (Optional but Recommended)

1. Go to **Users** → **Users**
 2. Click **Admin**
 3. Change the **Alias** to something non-obvious (e.g., `zbxadmin`)
 4. Save
-

2. Install Zabbix Agent 2 on Windows Hosts

Zabbix Agent 2 must be installed on every Windows server you want to monitor.

Download

Download the latest **Zabbix Agent 2** MSI for Windows from:

```
https://www.zabbix.com/download_agents
```

Select: **Version 7.0 LTS** → **Windows** → **64-bit** → **MSI**

Install via PowerShell (run as Administrator)

```
# Replace values in angle brackets
$ZabbixServer = "<ZABBIX_SERVER_IP>" # e.g. 54.83.1.183
$HostName     = "<THIS_SERVER_NAME>" # e.g. srv-webapps-01
```

```
msiexec /i "zabbix_agent2-7.0.0-windows-amd64.msi" /qn `
SERVER=$ZabbixServer `
SERVERACTIVE=$ZabbixServer `
HOSTNAME=$HostName
```

Verify the Service is Running

```
Get-Service "Zabbix Agent 2"
```

Expected output: Status: Running

Firewall Rule (if Windows Firewall is enabled)

```
New-NetFirewallRule -DisplayName "Zabbix Agent" `
-Direction Inbound -Protocol TCP -LocalPort 10050 `
-Action Allow
```

Config File Location

```
C:\Program Files\Zabbix Agent 2\zabbix_agent2.conf
```

Key settings to verify:

```
Server=<ZABBIX_SERVER_IP>
ServerActive=<ZABBIX_SERVER_IP>
Hostname=<THIS_SERVER_NAME>
```

After any config change, restart the service:

```
Restart-Service "Zabbix Agent 2"
```

3. Install Zabbix Agent 2 on Linux Hosts

```
wget https://repo.zabbix.com/zabbix/7.0/ubuntu/pool/main/z/zabbix-release/zabbix-
release_latest+ubuntu22.04_all.deb
dpkg -i zabbix-release_latest+ubuntu22.04_all.deb
apt-get update
apt-get install -y zabbix-agent2

# Set server IP and hostname
ZABBIX_SERVER="<ZABBIX_SERVER_IP>"
HOST_NAME="<THIS_SERVER_NAME>"

sed -i "s/^Server=.*Server=$ZABBIX_SERVER/" /etc/zabbix/zabbix_agent2.conf
sed -i "s/^ServerActive=.*ServerActive=$ZABBIX_SERVER/" /etc/zabbix/zabbix_agent2.conf
sed -i "s/^Hostname=.*Hostname=$HOST_NAME/" /etc/zabbix/zabbix_agent2.conf
```

```
systemctl enable zabbix-agent2
systemctl start zabbix-agent2
```

4. Add a Host in Zabbix

Steps

1. Go to **Monitoring** → **Hosts** (or **Configuration** → **Hosts** in older UI)
2. Click **Create host** (top right)
3. Fill in the **Host** tab:

Field	Value
Host name	Must match the Hostname= in the agent config exactly
Visible name	Friendly name (e.g. Web Server 01)
Templates	Add templates here (see Section 5)
Host groups	Select or create a group (e.g. Windows Servers, SQL Servers)

4. Click **Add** under **Interfaces** and choose **Agent**:

Field	Value
IP address	The host's IP address
DNS	Leave blank unless using DNS
Connect to	IP
Port	10050

5. Click **Add** to save the host

Active vs Passive Agent Mode

Mode	How it works	When to use
Passive (default)	Zabbix server connects to agent on port 10050	Host is directly reachable from Zabbix server
Active	Agent connects to Zabbix server on port 10051	Host is behind NAT/firewall (e.g. on-prem servers reaching Zabbix in AWS)

For on-prem servers connecting to Zabbix in AWS, use **Active** mode:

- Set `ServerActive=<ZABBIX_SERVER_IP>` in the agent config
- In the host interface, set the port to `10051` and select **Active checks only**

5. Apply Monitoring Templates

Templates define what gets monitored. Apply them on the **Host** tab when creating or editing a host.

Recommended Templates by Host Type

Windows Servers (IIS, general)

Template	Monitors
Windows by Zabbix agent	CPU, memory, disk, network, services, uptime
Windows services by Zabbix agent	Specific Windows services (configurable)
IIS by Zabbix agent	IIS worker processes, request queue, connections

SQL Server

Template	Monitors
Microsoft SQL Server by Zabbix agent	Connections, blocked processes, deadlocks, buffer cache hit ratio, query times

Linux Servers

Template	Monitors
Linux by Zabbix agent	CPU, memory, disk, network, load average

Network Devices (SNMP)

Template	Monitors
Network generic device by SNMP	Interfaces, bandwidth, errors
Cisco IOS by SNMP	For Cisco routers/switches

How to Apply a Template

1. Edit the host → **Templates** tab
 2. Start typing the template name in the search box
 3. Select and click **Add**
 4. Click **Update**
-

6. Verify the Host is Being Monitored

Check Host Status

1. Go to **Monitoring** → **Hosts**
2. The **Availability** column shows green (ZBX) when the agent is reachable
3. If red — the Zabbix server cannot reach the agent (check firewall/IP)

Check Latest Data

1. Click the host name → **Latest data**
2. You should see metrics populating within 1–2 minutes of adding the host

Test Agent Connectivity from the Zabbix Server

SSH into the Zabbix server and run:

```
zabbix_get -s <HOST_IP> -p 10050 -k agent.ping
```

Expected response: 1

If it times out — the agent is unreachable. Check:

- Agent is running on the host
- Port 10050 is open on the host's firewall
- Security group / network ACL allows traffic from the Zabbix server IP

7. Configure Email Alerting

Step 1 — Create a Media Type (SMTP)

1. Go to **Alerts** → **Media types**
2. Click **Email** (built-in)
3. Configure:

Field	Value
SMTP server	email-smtp.us-east-1.amazonaws.com (AWS SES)
SMTP port	587
SMTP helo	Your domain
SMTP email	Verified SES sender address
Connection security	STARTTLS
Authentication	Username and password
Username	SES SMTP username
Password	SES SMTP password

4. Click **Update**

Step 2 — Add Email to Your User

1. Go to **Users** → **Users** → click your admin user
2. Click **Media** tab → **Add**
3. Select **Email**, enter your email address
4. Set severity levels (check all: Disaster, High, Average, Warning, Information, Not classified)
5. Save

Step 3 — Create an Action

1. Go to **Alerts** → **Actions** → **Trigger actions**
2. Click **Create action**
3. Name it (e.g. Notify Admin)
4. Under **Operations**, click **Add**:
 - Operation type: Send message
 - Send to users: your admin user
 - Send only to: Email

5. Save

8. Recommended Global Settings

Go to **Administration** → **General** and review:

Setting	Recommended Value
Working time	1-7,00:00-24:00 (24/7)
Default severity	Adjust trigger colors to your preference
History storage period	90d
Trend storage period	365d

Create Host Groups (Organization)

Go to **Configuration** → **Host groups** and create groups before adding hosts:

- Windows Servers
- SQL Servers
- Linux Servers
- Network Devices
- AWS

9. Host Reference — BEL Infrastructure

Hosts to onboard based on current BEL infrastructure:

Hostname	IP	OS	Template(s)	Group
srv-webapps-01	TBD	Windows Server	Windows by Zabbix agent, IIS by Zabbix agent	Windows Servers
srv-webapps	TBD	Windows Server	Windows by Zabbix agent, IIS by Zabbix agent	Windows Servers
BEL-API1	TBD	Windows Server	Windows by Zabbix agent, IIS by Zabbix agent	Windows Servers
srv-sql01	TBD	Windows Server	Windows by Zabbix agent, Microsoft SQL Server by Zabbix agent	SQL Servers
SRV-Attunity02	TBD	Windows Server	Windows by Zabbix agent	Windows Servers
SRV-UTIL	192.168.1.61	Windows 2012 R2	Windows by Zabbix agent	Windows Servers

Note: On-prem servers connect to Zabbix in AWS via VPN. Use **Active** agent mode so agents initiate the connection outbound to the Zabbix server — no inbound firewall changes needed on-prem.

Agent Config for On-Prem Servers (Active Mode)

```
# C:\Program Files\Zabbix Agent 2\zabbix_agent2.conf
Server=54.83.1.183
ServerActive=54.83.1.183
Hostname=srv-webapps-01      # change per host
```

Quick Reference

Task	Location in UI
Add a host	Monitoring → Hosts → Create host
Apply a template	Host editor → Templates tab
Check agent status	Monitoring → Hosts → Availability column
View latest metrics	Monitoring → Hosts → Latest data
Configure email	Alerts → Media types → Email
Create alert action	Alerts → Actions → Trigger actions
View active problems	Monitoring → Problems
View dashboards	Monitoring → Dashboard